5 (a) Faraday's law of electromagnetic induction states that the magnitude of the induced emf is directly **proportional** to the rate of change of magnetic flux **linkage** or the rate of cutting of magnetic flux. A1

Comments:
Definitions must be phrased as the way it is. More importantly, e.m.f stands for electromotive force NOT electromagnetic force!!

(b) When the card is brought in range of/ near to a card reader, the **varying magnetic field** generated by the card reader results in a **changing magnetic flux linkage** through the loops/coils of printed wire embedded in the card. A1

By Faraday's law, there is an **emf induced in the loops**. (The emf is produced in a direction such that is opposes the varying magnetic field.) A1

The **closed circuit** allows the emf to drive a current and thus supplies power to the internal transmitter (which can then send its data to the card reader). A1

Comments:
Students should answer such that the "cause and effect" are clear from the marker's point of view.
Firstly, it is the changing magnetic flux linkage through the 3 loops that causes an emf to be induced.
Secondly, when implying there is induced emf, Faraday's law must be stated since the question said "using Faraday's law".
Thirdly, students have to explain why the transmitter receives electrical power by saying there is an induced current flowing since circuit is closed.

(c) (i)

$$\varepsilon = -\frac{d\Phi}{dt} = -\frac{d(NBA\sin 2\pi ft)}{dt}$$

$$= -2\pi fNBA\cos 2\pi ft \qquad \text{M1}$$

Magnitude of peak emf = ε_0 when $\cos 2\pi ft = 1$ M1

$$\varepsilon_0 = 2\pi fNAB_0$$

$$= 2\pi(13.6\times10^6)(3)(4.0\times10^{-3})B_0 \qquad \text{M1}$$

$$= 1.02\times10^6\ B_0 \qquad \text{A0}$$

(ii)

$$V_{rms} = \frac{V_0}{\sqrt{2}} \qquad \text{B1}$$

$$10.0 \times 10^{-3} = 1.02 \times 10^{6} \frac{B_0}{\sqrt{2}}$$ A1

$$B_0 = 1.38 \times 10^{-8} \text{ T}$$

<u>Comments</u>:
The question asks "to show clearly" hence working must be detailed with good physics shown.

(d) The card is designed to operate if a current flows through it, regardless of direction. Since when tapped with either face, the changing magnetic flux linkage through the loops of wire will be able to induce an emf and allow current to flow to power the transmitter. A1

<u>Comments</u>:
A lot of answers are unclear as they stopped short of explaining induced current will flow regardless which face of the card is tapped.